Batangas State University
College of Informatics and Computing Sciences

SYSTEM PROPOSAL

CafeTrack: Real-Time Stock Monitoring with QR-Based Authentication for Merrylane Cafe Foodhub

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INTRODUCTION

Merrylane Cafe Foodhub is a micro-enterprise located in Lipa City, Batangas. It operates in the food service industry, serving coffee, beverages, and light meals to walk-in customers, students, and nearby residents. The cafe functions as a single-branch business managed by an owner-manager who oversees day-to-day operations, with a small team of cooks and baristas handling preparation and service.

In terms of scale, the cafe falls under the category of a micro-enterprise under Philippine MSME classification. Its ingredients are sourced from local suppliers and stored on-site in a compact storage area that holds both perishable items (dairy, produce, syrups) and non-perishable goods (dry mixes, packaged ingredients). The cafe operates on tight margins, which makes inventory accuracy and waste control directly tied to profitability.

The technological environment of the cafe is modest but sufficient. The business has access to a working computer or laptop, a smartphone, and an internet connection. It does not currently use any dedicated inventory management software. Stock tracking is performed entirely by hand using a notebook. When ingredients arrive, the date and quantity are written down. When items are taken from storage for use, the notebook is updated again. A weekly physical count checks whether the notebook matches the shelves. This manual approach is the current state of the cafe's inventory system.

The cafe's existing computer is the intended deployment target for CafeTrack. The system runs locally by default, so daily operations do not depend on an internet connection. An optional cloud-backed configuration uses Supabase as the data layer for deployments where automatic off-site backups or remote dashboard access are needed.

PROBLEM IDENTIFICATION

Merrylane Cafe Foodhub currently tracks inventory through a handwritten notebook. This method creates four operational problems that affect daily operations, cost efficiency, and accountability. Each problem is significant enough that manual tracking can no longer address it, and each points to the need for an IT-based solution.

2.1 Expiration and Spoilage

Ingredients expire without anyone noticing until they are already spoiled. The cafe loses money on discarded items and risks using expired ingredients in food preparation. Because expiration dates are recorded by hand if they are recorded at all the owner has no reliable way to know which items should be prioritized for use. Balisi (2023) observed this same pattern in a study of a Philippine food service business, noting that spoilage is a direct consequence of informal record-keeping.

During our observation of a typical week, the cafe discarded a noticeable amount of dairy, produce, and perishable syrups because no one remembered to check the dates. Mishra et al. (2024) confirmed that smart inventory applications designed to track expiration dates and manage food supplies can significantly reduce waste in restaurants.

2.2 Reactive Restocking

The cafe only discovers that an ingredient is out of stock when someone needs it. This leads to rushed purchases, delayed orders, and menu items that cannot be served. There is no way to know in advance that an item is running low, so reorders happen reactively rather than proactively. During our observation, staff at the cafe discovered at the peak of service that a key ingredient had run out, and the only option was an emergency trip to the nearest supplier. This pattern repeats on busy weekends and creates unnecessary pressure during peak hours. Tong (2025) showed that real-time barcode tracking provides the data needed for proactive ordering, but the cafe has no such system in place.

2.3 Slow and Error-Prone Checkout

The manual checkout process is slow and prone to error. During peak hours, staff spend too much time writing down items taken from storage. Transcription errors cause stock records to drift from reality. Multi-item transactions are especially slow and easy to get wrong. Fauziah et al. (2023) documented the same problem in a cafe setting, finding that manual stock collection led to

calculation errors and long delays in checking item quantities. Ahrens and Chapman (2004) found that successful restaurants combine standardized processes with flexibility during peak periods, but manual tracking forces staff to choose between speed and accuracy. A single checkout on a busy morning at the cafe can take thirty seconds or more, and mistakes are usually not discovered until the next physical count, by which point the source of the error is impossible to trace.

2.4 No Accountability for Stock Movements

There is no accountability for stock movements. When stock discrepancies occur, the cafe cannot determine who took what, when, or why. This creates friction among staff and makes it difficult for the owner to enforce basic inventory discipline. Without a reliable record, the owner has to rely on memory and trust, which is neither scalable nor fair to staff who follow the rules. Balisi (2023) noted the same issue, finding that informal record-keeping undermines accountability in small food businesses.

2.5 Common Cause and Significance

Taken together, these four problems share a common cause: the cafe has no real-time, tamper-proof record of what is in storage, what is being used, and what is running low or about to expire. A manual notebook cannot deliver that. The problem is significant because it directly affects the cafe's profitability through waste, missed sales, and untraceable losses. It also affects customer experience through slower service and occasional unavailability of menu items. Finally, it affects workplace trust because discrepancies cannot be fairly resolved.

An IT-based solution is required because the volume and speed of inventory movement exceed what handwriting can reliably capture during peak service. A system that scans barcodes, updates stock in real time, fires automated alerts, and logs every transaction with a user and a timestamp can address all four problems with a single, integrated tool.

PROJECT OBJECTIVES

To develop a connected inventory management system that allows administrators and staff to manually encode and generate barcodes for ingredients, while enabling staff to perform quick, barcode-based checkout, restock, and waste disposal transactions with real-time stock deduction for Merrylane Cafe Foodhub.

Specifically, the project aims to:

1. **Create a web-based dashboard** for administrators and staff to manually add, update, and manage inventory items, and automatically generate unique barcodes for each ingredient.
2. **Develop a secure, barcode-based transaction process** where staff authenticate via personal barcodes and scan item barcodes to accurately record checkout, restock, and waste disposal transactions.
3. **Ensure seamless data synchronization** between the physical barcode scanners and the central database, providing the admin with real-time stock visibility and updates.
4. **Integrate automated alerts** for low-stock levels and near-expiry items to proactively minimize ingredient waste and spoilage.
5. **Maintain a secure, immutable audit trail** that logs every transaction, including item additions, checkouts, restocks, and waste disposal, with user identification and timestamps for strict accountability.

PROPOSED IT SOLUTION

CafeTrack is a web-based inventory management system that links physical ingredients to digital records through barcodes. Both the admin and staff can register new ingredients directly through the dashboard, and the system generates a unique barcode for each item as soon as it is saved. During daily operations, staff scan these barcodes to either check out items for use, restock items when deliveries arrive, or record items disposed of as waste. Each scan updates stock levels in real time. Automated alerts fire when items are running low or close to expiration. Every transaction is logged with user identification and a timestamp, giving the owner a complete, unalterable record of what moved, when, and by whom.

4.1 System Features

- **Role-Based Access Control (RBAC):** Distinct permissions for Admins (owner/manager) who manage the system, and Staff (cooks/baristas) who perform checkout, restock, and waste disposal operations. Both roles can add new items to the system.
- **Barcode Authentication (Staff Login):** Staff scan their personal barcode to log in and enter a password to verify their identity before performing any inventory transactions.
- **Manual Item Management (Admin and Staff):** Both admins and staff can manually add, edit, or delete ingredients through the web dashboard, inputting details such as

name, category, physical form, quantity, expiration date, low-stock threshold, and storage location.

- **Automatic Barcode Generation:** The system automatically generates unique barcodes for each ingredient upon manual entry. Barcodes can be previewed on screen and printed on durable labels for storage containers.
- **Barcode-Based Item Transactions:** Staff can scan item barcodes using connected scanner devices to record three transaction types checkout (items taken for use), restock (items added back to inventory), and waste disposal (items discarded due to spoilage or damage).
- **Multi-Item Transactions with Confirmation:** Staff can scan multiple items in a single transaction, with a final confirmation step (re-entering password or PIN) required to prevent accidental entries.
- **Real-Time Stock Updates:** Inventory levels automatically adjust in the database and reflect on the admin dashboard immediately after every transaction.
- **Automated Alerts and Reports:** The system generates real-time alerts for low-stock levels and near-expiry items, plus historical reports for inventory usage and waste tracking. Reports are exportable to PDF or CSV.
- **Immutable Audit Trail:** A secure, append-only chronological log records all system activities, item additions, checkouts, restocks, waste disposals, adjustments, logins, and failed logins with user identification, timestamp, and outcome.
- **Expiration and Out-of-Stock Alert:** The system alerts the admin about items that are near expiration, out of stock, or already expired.

4.2 Proposed Additional Features

Beyond the original scope, the following features are proposed because the cafe's actual operations call for them.

- **Per-Box Packaging and Restocking:** Ingredients will be definable with a box size, so a restock can be entered as a box count rather than a unit count. The database will perform the conversion automatically.
- **Minimum Shelf Life on Arrival:** A storage location will be configurable with a minimum remaining shelf life. When a restock arrives with an expiration date too close to the threshold, the system will flag the delivery for review.
- **Staff Submission with Approval Workflow:** Staff without item-creation privileges will be able to submit a new ingredient for approval. The item will not be added to stock until an admin reviews and approves it.

- **Email Notifications:** Optional email alerts for low stock, out of stock, near-expiry items, expired items, daily summaries, and rejected sign-in attempts. Off by default; an admin will enable them.
- **Automatic Daily Backup:** The system will back up the database daily after 3:00 AM, retain the three most recent backups, and run without taking the system offline.
- **Cloud Data Layer (Optional):** The system will run on a local embedded PostgreSQL database (PGlite) by default. Setting a single environment variable will switch the data layer to Supabase.

4.3 Intended Users

User	Role	Primary Activities
Owner-Manager	Admin	Monitors stock levels, manages users, reviews alerts and reports, audits transactions, approves staff submissions, configures system settings, enables email notifications.
Cooks and Baristas	Staff	Scans personal barcodes to log in, scans item barcodes to perform checkout, restock, and waste disposal transactions, adds new ingredients when deliveries arrive, submits new item requests for approval.

4.4 Scope and Delimitations

The system covers item management with both admin and staff able to add items, barcode generation, barcode-based authentication, three types of barcode transactions (checkout, restock, and waste disposal), per-box packaging, minimum shelf life enforcement, real-time stock updates, automated alerts, email notifications, staff submission with approval, an immutable audit trail, automatic daily backups, and usage reports. It is designed for a single branch of Merrylane Cafe Foodhub and runs on the cafe's existing computer with a standard USB barcode scanner.

The following are outside the scope of this project: payroll integration, POS integration, online ordering, supplier management, multi-branch support, BIR compliance, and full offline operation. The system runs locally and does not require an internet connection for daily use, but the optional cloud-backed configuration depends on connectivity to reach Supabase. Out-of-scope items may be considered for future development.

4.5 Requirement Analysis

The requirements below were gathered through the client interview and refined during the design phase. They define what the system must do (functional requirements) and how well it must perform (non-functional requirements). Each functional requirement is assigned a code for traceability during development and testing.

Functional requirements

Code	Module	Description
FR-01	User Authentication	The system shall allow staff to log in by scanning their personal barcode and entering a password. Admins shall log in using username and password. Failed sign-ins are recorded in the audit log.
FR-02	User Management	The system shall allow the admin to add, edit, disable, and enable user accounts, assign roles (Admin or Staff), generate personal barcode credentials for each staff member, rotate a staff member's barcode (invalidating any previously printed label), and deactivate an account with an optional reason that will be shown to the user when they attempt to sign in.
FR-03	Item Management	The system shall allow both admin and staff to add, edit, or delete ingredients through the dashboard, entering details such as name, category, physical form, quantity, expiration date, and storage location. Staff without item-creation privileges can submit new items for admin approval.
FR-04	Barcode Generation	The system shall automatically generate a unique barcode for each ingredient upon item creation. The barcode can be previewed on screen and printed on durable labels for storage containers.
FR-05	Barcode Transactions	The system shall allow staff to scan item barcodes to record three types of transactions: checkout, restock, and waste disposal. Restocks can be entered as a box count and are converted to units automatically.
FR-06	Multi-Item Checkout	The system shall allow staff to scan multiple items in a single transaction. A final confirmation step (re-entering password or PIN) shall be required before the transaction is committed.

FR-07	Real-Time Stock Updates	The system shall update inventory levels in the database immediately after every transaction and reflect the change on the admin dashboard without requiring a manual page refresh.
FR-08	Automated Alerts	The system shall generate real-time alerts for low-stock items, near-expiry items, out-of-stock items, and expired items. Alerts shall be displayed on the admin dashboard and sent by email if enabled.
FR-09	Reports	The system shall generate inventory usage reports and waste tracking reports, with filters for date range, category, and user. Reports shall be exportable to PDF or CSV.
FR-10	Audit Trail	The system shall log every transaction and system action (item additions, checkouts, restocks, waste disposals, adjustments, logins, and failed logins) with user identification, timestamp, and outcome. The audit log shall be append-only. The admin shall be able to verify the integrity of the hash chain from the audit page.
FR-11	Email Integration	The system shall send email alerts to the admin for low-stock, out-of-stock, near-expiry, and expired items, daily summaries, and failed login warnings. The admin shall be able to configure recipients per alert type, configure the sending account, run a test send, and view a delivery log of recently sent messages.
FR-12	Backup	The system shall perform an automatic daily backup after 3:00 AM, retain the three most recent backups, and operate without taking the system offline.

Non-Functional Requirements

Code	Module	Description
FR-01	Usability	The interface shall be simple enough that staff can learn to use the system with minimal training. Role-based navigation shall show only the modules each user is authorized to access.
NFR-02	Security	The system shall enforce role-based access control. Staff authentication shall require both a personal barcode scan and a password. All data in transit shall be encrypted using TLS 1.3. All stored credentials shall be hashed.
NFR-03	Performance	Barcode scan processing shall complete in under 1 second. The admin dashboard shall reflect stock changes within 2 seconds. Item lookup queries shall complete in under 100 milliseconds.
NFR-04	Reliability	The system shall target 99.5% uptime during operating hours. Data shall persist across sessions. The system shall prevent negative stock through optimistic locking on stock deduction. Automated backups shall run daily.
NFR-05	Scalability	The system shall support multiple item categories and storage locations, handle at least 500 items and thousands of transactions without performance loss, and use a modular architecture that supports future feature expansion.
NFR-06	Maintainability	The system shall follow a modular architecture with a clear separation of concerns (InventoryService, AuthService, ReportService). All code shall be documented and follow consistent naming conventions.

4.6 Email Integration (FR-11)

Alerts currently only show on the dashboard. With email integration, the admin will receive alerts in their inbox even when not looking at the system. This supports the goal of minimizing waste by shortening the time between a stock event and a response.

- **Low-stock alert** — when an item hits its reorder level.
- **Near-expiry alert** — daily for items expiring within 3 days.
- **Out-of-stock alert** — immediately when stock reaches zero.
- **Expired items** — daily list of items past their expiration date.
- **Daily summary** — end-of-day totals, low stock, expiring items, transactions.
- **Failed login warning** — after 3 failed attempts in 10 minutes.
- **Reports by email** — admin can send any report to an email address.
- **Recipient settings** — admin can add, remove, or toggle recipients per alert type.

4.7 Administrative Controls and System Configuration

Beyond item and transaction management, CafeTrack will provide the admin with a set of configuration and control features that govern how the system behaves, how staff credentials are managed, and how policy thresholds are tuned to the cafe's reality. These are gathered on the *Users*, *Settings*, *Email*, and *Audit* pages, all of which are admin-restricted.

Staff Account Lifecycle Controls

- **Barcode rotation:** The admin can regenerate a staff member's personal barcode at any time. Rotation invalidates any previously printed label immediately, so a lost or leaked barcode can be voided in one action. The system confirms the consequence before performing the rotation.
- **Approval code management:** Each staff member can be issued a short approval code that the admin reads out and hands over in person. The code authorizes the staff member to submit new items for approval. The admin can issue or revoke the code; the code's value is visible to the admin while active.
- **Deactivation with reason:** An admin can deactivate a staff account and optionally record a reason (up to 500 characters). The reason is shown to the user on their next sign-in attempt, so the decision is communicated rather than silent. Reactivation restores access in one action.
- **Per-staff item management permission:** In addition to the Admin/Staff role, the admin can grant or revoke the ability for an individual staff member to create items directly. This is a reversible, audited toggle, so a trusted barista can be permitted to add deliveries without being promoted to full Admin.

Policy and Reference Data Settings

- **Expiry warning window:** The number of days before expiration at which a "near-expiry" alert fires (configurable from 1 to 365 days). This lets the admin tune the alert horizon to the

shelf life of the cafe's actual inventory.

- **Session timeout:** The idle period after which a signed-in user is automatically signed out (configurable from 1 to 480 minutes). This limits exposure if a shared terminal is left unattended.
- **Business name:** The name that appears on reports, exports, and printed labels.
- **Reference list management:** The admin can add and remove the values that appear in the Category and Storage Location dropdowns. Deleting a value that is still assigned to an item is blocked by the server with a clear message, so historical records never point at a category that no longer exists. Each storage location also carries an inline minimum shelf life value, which the admin can set or clear.

Audit Integrity Verification

- **Verify chain:** The audit page provides a one-click action that recomputes and verifies the hash chain across all log entries. The result is shown as a persistent banner: either confirmation that no entry has been altered, or a list of the specific entries where the chain does not match. This turns the tamper-evident audit log from a claim into a check the owner can run at any time.

Email Operational Controls

- **Setup gate:** Email alerts remain disabled until the sending account has been configured and verified. The server enforces the same gate, so the switch cannot be bypassed by calling the API directly.
- **Combined save-and-test:** The admin saves the sending account and immediately triggers a test message. If the test fails, the failure is shown as a persistent panel with the SMTP provider's exact error text, so the admin can act on the provider's own diagnosis rather than a generic message.
- **Delivery log:** A recent history of every alert email the system has sent, showing the recipient, subject, timestamp, and outcome, so the admin can confirm that a critical alert actually reached its destination.

SYSTEM DESIGN AND METHODOLOGY

5.1 Development Methodology

The project will follow an Agile development methodology organized into seven weekly sprints. Agile was chosen because it fits the reality of building a system for a small business. Requirements evolve as the client sees working software, and feedback from actual users is more useful than an exhaustive upfront specification. Instead of committing to a fixed plan and delivering

everything at the end, the team will deliver working increments of CafeTrack every week and adjust the next sprint based on what the client learns from the previous one.

Each sprint will follow a consistent rhythm. On the first day, the team will meet with Merrylane Cafe Foodhub management to confirm what will be built and to raise any new requirements from the previous sprint. During development days, the team will build and test. Mid-sprint, the team will hold a short internal review to catch blockers early. At the end of each sprint, the team will demonstrate the increment to the client, collect feedback, and reflect on what went well and what could be improved. The next sprint will start with a clear understanding of what to prioritize.

5.2 System Architecture

CafeTrack will use a three-tier architecture optimized for local-first deployment. The presentation layer is a web-based dashboard accessible through a standard browser, built with HTML, CSS, and JavaScript. The application layer handles business logic authentication, checkout processing, alert generation, barcode printing, and report compilation and runs locally on the cafe's existing computer. The data layer defaults to PGlite, an embedded PostgreSQL engine that stores all data in a local file on the same machine.

This local-first design was chosen to keep the critical path scan, validate, and confirm independent of the network. Barcode processing will complete in milliseconds regardless of internet state, so a brief connectivity issue will not stop service. No external service will be required for daily operation.

For deployments that need automatic off-site backups or remote dashboard access, the data layer can be switched to Supabase by setting a single environment variable. The same application code runs against either data layer, so no feature differences exist between the two deployments.

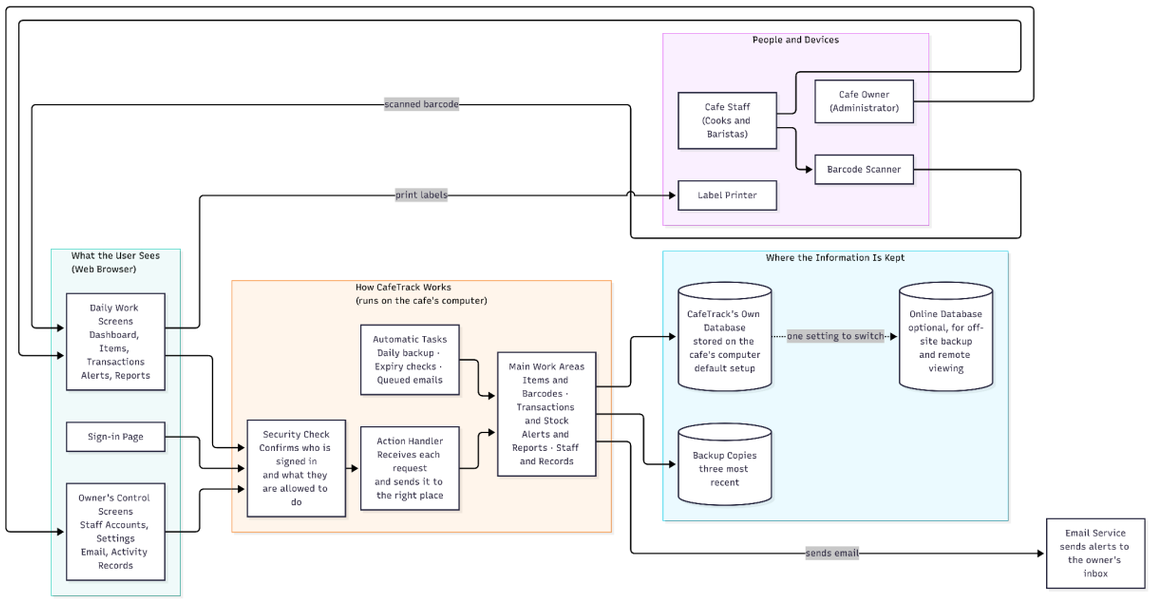


Figure 1. System Architecture of CafeTrack

5.3 Data Flow Diagram

The Data Flow Diagram (DFD) shows how information moves through CafeTrack from the external entities that interact with the system, through the processes that handle that data, and into the data stores where records are kept. Two levels of DFD are presented: a context diagram that shows the system as a whole, and a level 1 diagram that breaks it down into its main processes.

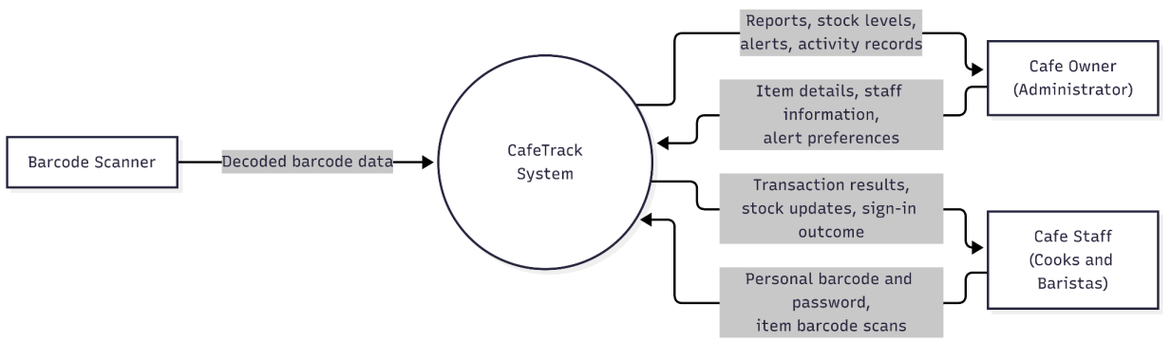


Figure 2. Context Diagram of CafeTrack

The context diagram shows CafeTrack as a single process that interacts with three external entities. The Admin provides item details, user information, and alert settings, and in return receives reports, stock levels, and alerts. Staff provide their personal barcode and password to log in, then scan item barcodes during checkout, restock, or waste disposal. The barcode scanner is treated as a separate external entity because it captures physical input and passes decoded data to the system.

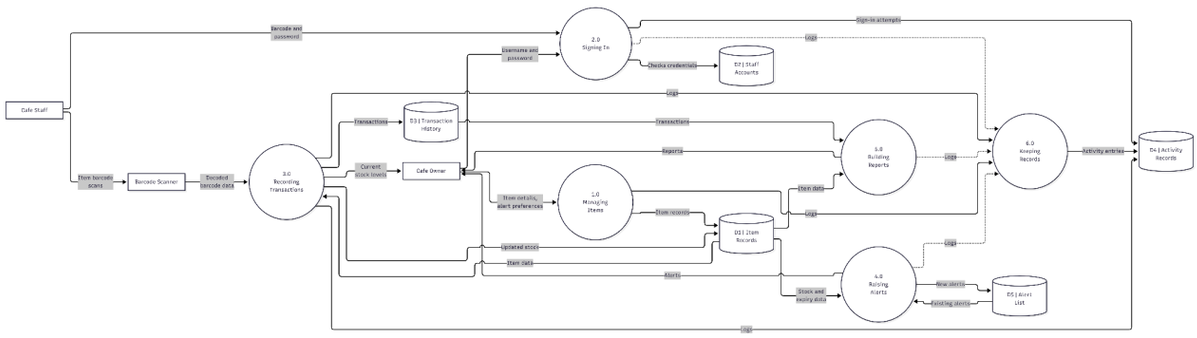


Figure 3. Level 1 Data Flow Diagram of Cafetrack

The level 1 diagram breaks the system into six processes. Process 1 handles all item management and writes to the item records store. Process 2 validates credentials and writes login attempts to the audit trail. Process 3 is the core transaction handler it reads item records, checks available stock, updates the item quantity, and writes to both the transaction log and audit trail. Process 4 reads item records and alerts to check for low stock or near-expiry conditions, and writes new alerts back to the alert store. Process 5 reads the transaction log and item records to build reports. Process 6 is a cross-cutting process that is invoked by every other process to log actions to the audit trail. The five data stores separate operational data (D1-D3, D5) from the immutable audit history (D4).

5.4 Entity Relationship Diagram

The Entity Relationship Diagram (ERD) shows the main entities that store data in CafeTrack and the relationships between them. It is the reference for the database schema.

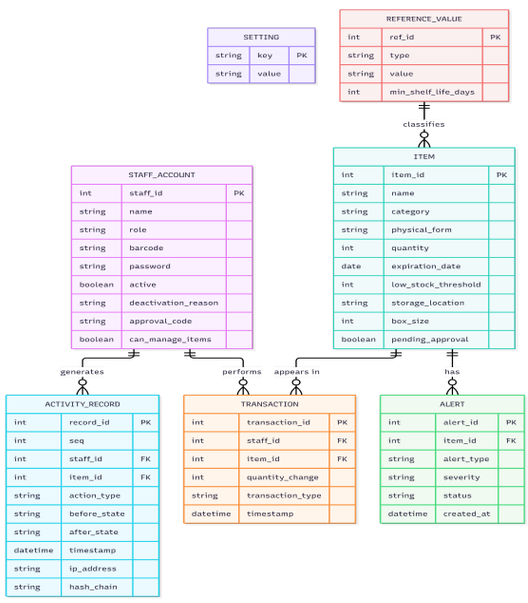


figure 4. Entity Relationship Diagram of CafeTrack

Entities and Attributes

- **USER** (user_id PK, name, role, barcode, password_hash, active, deactivation_reason, approval_code, can_manage_items)
- **ITEM** (item_id PK, name, category, physical_form, quantity, expiration_date, low_stock_threshold, storage_location, box_size, pending_approval)
- **TRANSACTION** (transaction_id PK, user_id FK, item_id FK, quantity_change, transaction_type, timestamp)
- **AUDIT_LOG** (log_id PK, seq, user_id FK, action_type, item_id FK, before_state, after_state, timestamp, ip_address, hash_chain)
- **ALERT** (alert_id PK, item_id FK, alert_type, severity, status, created_at)
- **SETTING** (key PK, value) stores business name, expiry warning window, session timeout, and email configuration
- **REFERENCE_VALUE** (ref_id PK, type, value, min_shelf_life_days) backs the Category and Storage Location lists

Relationships

- One USER can perform many TRANSACTIONS (1:N)
- One ITEM can appear in many TRANSACTIONS (1:N)
- One USER can generate many AUDIT_LOG entries (1:N)
- One ITEM can have many ALERTS over time (1:N)
- Reference values of type *category* and *storage location* are referenced by ITEM (1:N)

5.5 Wireframes

The following wireframes show the intended layout of the two primary screens: the login screen used by both admin and staff, and the inventory dashboard used by the admin. Both will be implemented in the build.

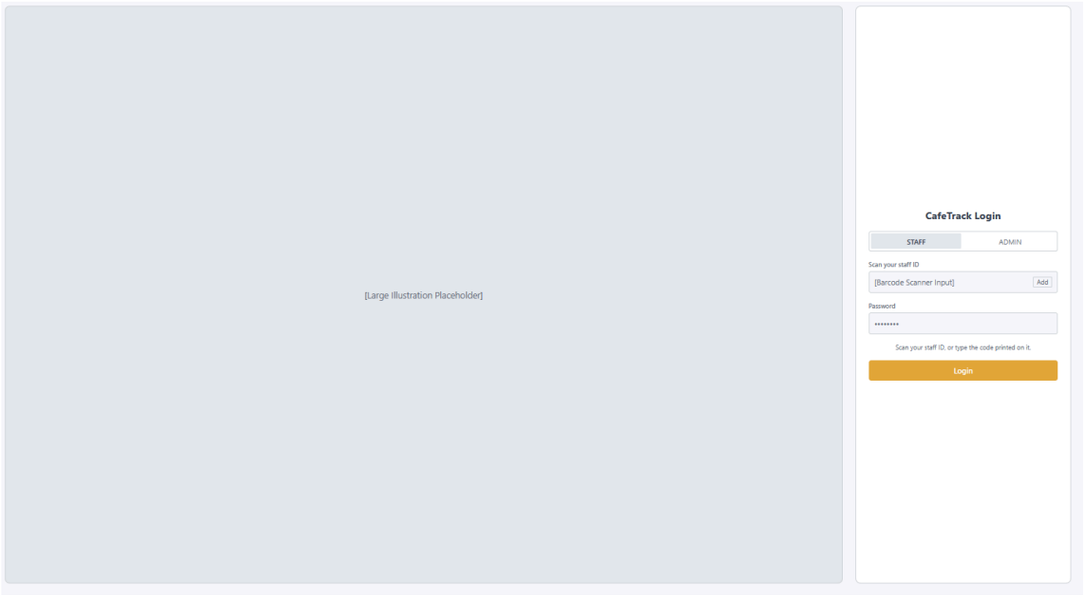


Figure 5. User Login Wireframe

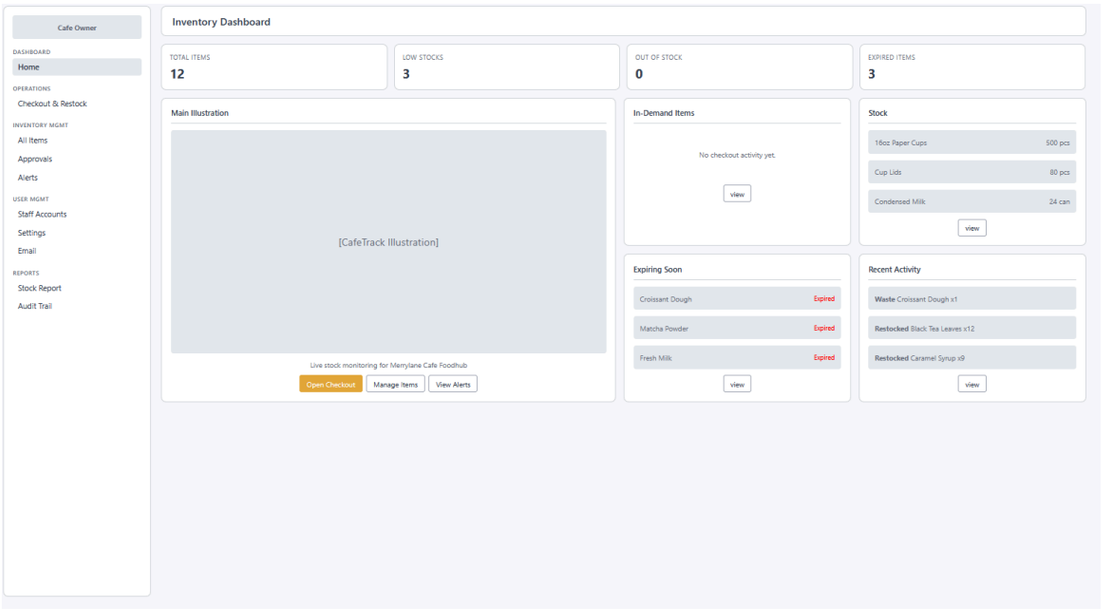


Figure 6. Inventory Dashboard Wireframe

5.6 Barcode Library Selection

To fulfill FR-04 (Automatic Barcode Generation), which requires the system to generate a unique barcode for each ingredient upon creation, preview it on screen, and print it on durable labels, CafeTrack requires a barcode generation library that runs both in the browser for preview and on the local Node.js server for label-sheet generation without introducing a network dependency.

JsBarcode was selected to fulfill this requirement (Lindell, n.d.). It is an open-source barcode generator distributed under the MIT license and written in JavaScript with no runtime dependencies, which keeps the deployment footprint small and the dependency tree auditable.

The library supports ten 1D barcode formats. For CafeTrack, CODE128 is used. CODE128 encodes the full ASCII character set, is self-checking via an internal checksum, and is recommended for inventory and asset tracking because it fits long alphanumeric identifiers into a narrow label without sacrificing print quality. Two classes of identifiers are encoded: item identifiers in the form ITEM-0042 and staff badge identifiers in the form STAFF-0007. Both fit comfortably within a CODE128 symbol at a label width of roughly 40 to 50 millimeters, which is the practical size for a sticker applied to a storage container or an identification card.

JsBarcode renders to SVG for on-screen preview inside the item form, to Canvas for composing multiple labels onto a single print sheet, and to Node.js output on the server for batch label generation. The API requires a single call to render a barcode into a targeted element, with options for width, height, line color, margin, and human-readable text passed as a plain object. No configuration file, initialization step, or build-time tooling is required, which keeps integration with FR-04 straightforward.

When an item is saved, the system assigns the next sequential identifier and renders the barcode preview inside the item detail panel. The print action serializes that same identifier through the label-sheet template so the printed sticker carries a barcode identical to the one shown on screen. When the admin rotates a staff badge, the system assigns a new identifier and immediately regenerates the barcode, so no reprint of the old label can resolve to an active account.

The selection of JsBarcode rests on four properties that match the project's constraints. First, it has zero runtime dependencies, which keeps the deployment footprint small and the dependency tree auditable. Second, it runs unchanged in the browser and in Node.js, so there is only one barcode implementation to maintain and one behavior to test. Third, CODE128 is supported natively as a first-class format and has been stable across multiple major versions. Fourth, the MIT license permits unrestricted use, redistribution, and modification, so the library can ship with the deployed system without additional legal consideration (Lindell, n.d.).

If a future version of CafeTrack adds a second symbology, the barcode generation module is isolated enough that an additional renderer can be introduced alongside JsBarcode without rewriting the existing label pipeline. For the current scope, JsBarcode alone is sufficient.

IMPLEMENTATION PLAN

The implementation will be structured into seven sprints spanning seven weeks, in line with the Agile methodology described in the System Design and Methodology section. Each sprint will produce a working increment that will be demonstrated to Merrylane Cafe Foodhub management before the next sprint begins.

6.1 Project Phases and Timeline

Week	Planned Task	Status	Planned Completion Date	Actual Completion Date
1	Requirements gathering and analysis; client interview with Merrylane Cafe Foodhub	Ongoing	October 7, 2026	—
2	System design: ERD, DFD, UI wireframes; Barcode library research	Done	October 8–10, 2026	Oct 11, 2026
3	Planning system execution; distributing tasks among members; setting up development environment	Done	October 11, 2026	Oct 11, 2026
4	Documentation defense; start of system development (item management, QR generation)	Not yet	October 12–14, 2026	—
5	System development phase (QR authentication,	Not yet	October 15, 2026 – November 17, 2026	—

	checkout/restock, real-time updates)			
6	Final system completion; developer testing; integration testing	Not yet	November 18, 2026 – November 30, 2026	—
7	Final project presentation and user acceptance testing	Not yet	December 1, 2026	—

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